American Computer Science League

2020-2021 ● Contest 1: Shorts ● Elementary Division

1. Computer Number Systems What is the base 16 equivalent for 41375_8?	A. $42FD_{16}$ B. $413FA_{16}$ C. $85F5_{16}$ D. $42DF_{16}$ E. 4300_{16}
2. Computer Number Systems Which of the following is the smallest number? 100101110_2 457_8 $12D_{16}$	A. 100101110_2 B. 457_8 C. $12D_{16}$ D. 2 of them are the smallest E. all 3 are the same
3. Computer Number Systems Evaluate: $6023_8 + 757_8 + 1757_8$ Express the answer in octal.	A. 8759_8 B. 10761_8 C. 10527_8 D. 1761_8 E. 10771_8
4. Computer Number Systems How many more 1's are there in the binary numbers from 26_{10} to 35_{10} than the binary numbers from 16_{10} to 25_{10}?	A. 4 B. 5 C. 6 D. 7 E. 8
5. Computer Number Systems What is the positive difference between the hexadecimal values of the blue components represented by the color values $25A8F6_{16}$ and $C2D9AB_{16}$?	A. $4B_{16}$ B. $2A_{16}$ C. $9D_{16}$ D. 45_{16} E. $5B_{16}$

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1. Computer Number Systems

如何用十六进制表示八进制数 41375_8?

A. $42FD_{16}$
B. $413FA_{16}$
C. $85F5_{16}$
D. $42DF_{16}$
E. 4300_{16}

2. Computer Number Systems

下列数值中最小的是?

100101110_2 457_8 $12D_{16}$

A. 100101110_2
B. 457_8
C. $12D_{16}$
D. 其中两个数值最小
E. 三个数值大小都一样

3. Computer Number Systems

计算 $6023_8 + 757_8 + 1757_8$
并将计算结果用八进制表示。

A. 8759_8
B. 10761_8
C. 10527_8
D. 1761_8
E. 10771_8

4. Computer Number Systems

把从 26_{10} 到 35_{10} 的十进制数和从 16_{10} 到 25_{10} 的十进制数分别转换为二进制的表示形式后，前者含有数字 1 的总个数与后者含有数字 1 的总个数相差多少?

A. 4
B. 5
C. 6
D. 7
E. 8

5. Computer Number Systems

由十六进制数 $25A8F6_{16}$ 和 $C2D9AB_{16}$ 表示的颜色值，其中代表蓝色部分的差值是多少?

A. $4B_{16}$
B. $2A_{16}$
C. $9D_{16}$
D. 45_{16}
E. $5B_{16}$